

Aditya Mohan

REINFORCEMENT LEARNING RESEARCHER | PHD CANDIDATE

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- 🏠 Learning structured representations for generalization in offline and zero-shot RL
- 🤖 Applications to robotic manipulation and cloud autoscaling
- ⚡ Research experience at Amazon AWS, UT Austin, and the University of Tübingen

Research Experience

Amazon AWS

RESEARCH SCIENTIST INTERN

Berlin, Germany

Aug 2025 - Feb 2026

- **Topic:** Offline RL for cloud autoscaling with adaptable objectives and utilities.
- **Manager:** Dr. Huibin Shen

Leibniz University Hannover

PHD STUDENT

Hannover, Germany

April 2022 - April 2027

- **Topic:** Generalization in Reinforcement Learning, Meta-Learning and Algorithm Configuration
- **Supervisor:** Prof. Marius Lindauer

Learning and Intelligent Systems Group

RESEARCH INTERN

Berlin, Germany

Oct 2020 - Dec 2020

- **Topic:** Plan-Conditioned Policies for Sample-Efficient RL
- **Supervisor:** Ingmar Schubert, Prof. Marc Toussaint

Research Visits

May 2026 - **Prof. Amy Zhang at University of Texas at Austin**, Structured representations for zero-shot RL in
Aug 2026 robotic manipulation.

Austin, TX, USA

Feb 2025 - **Prof. Georg Martius at University of Tübingen**, Action-free representations for Zero-shot
Apr 2025 Reinforcement Learning

Tübingen, Germany

Jul 2023, **Dr. Tome Eftimov at Jožef Stefan Institute**, Dynamic Algorithm Configuration using RL
Jul 2024

Ljubljana, Slovenia

Education

Technical University of Berlin & EURECOM

M.SC. IN AUTONOMOUS SYSTEMS

Berlin, Germany

2019 - 2021

- **Thesis:** RL agents that quickly adapt to a partner for Ad-Hoc cooperation in the game of Hanabi (Grade 1.0)
- **Supervisor:** Prof. Klaus Obermayer

Manipal Institute of Technology

B.TECH. IN ELECTRONICS AND COMMUNICATION ENGINEERING

Karnataka, India

2013 - 2016

- **Thesis:** Development of Software for Autonomous Driving Support (Grade 1.0)
- **Supervisor:** Dr. S. Bhat M

Publications

🔗 Google Scholar 📄 DBLP 🆔 0000-0001-5561-5908

citations 409 – h-index 8 – i10 index 8

Journal & Conference Publications

- 1 Jan Malte Töpperwien*, **Aditya Mohan***, and Marius Lindauer. “Beyond Success Rates: Trainability and Extractability in Offline GCRL”. In: **NeurIPS 2026, Evaluations and Datasets Track**. Oct. 2026.
- 2 **Aditya Mohan***, Theresa Eimer*, Carolin Benjamins, Marius Lindauer, and Andre Biedenkapp. “Mighty: A Comprehensive Tool for studying Generalization, Meta-RL and AutoRL”. In: **JOSS**. Aug. 2026.

- 3 Lukas Koenemann, **Aditya Mohan**, Astrid Bensmann, Johannes Gerster, Marius Lindauer, and Richard Hanke-Rauschenbach. "Reinforcement learning-based dispatch for decentralized energy systems: Direct comparison of actor-critic method and neuroevolutionary approach". In: **Energy Conversion and Management: X** (July 2026).
- 4 Jannis Becktepe, Julian Dierkes, Carolin Benjamins, **Aditya Mohan**, David Salinas, Raghu Rajan, Frank Hutter, Holger Hoos, Marius Lindauer, and Theresa Eimer. "ARLBench: Flexible and Efficient Benchmarking for Hyperparameter Optimization in Reinforcement Learning". In: **Journal of Data-centric Machine Learning Research (DMLR)** (Feb. 2026).
- 5 **Aditya Mohan**, Amy Zhang, and Marius Lindauer. "Structure in Deep Reinforcement Learning: A Survey and Open Problems". In: **JAIR**. 2024.
- 6 Carolin Benjamins, Georgina Cenikj, Ana Nikolij, **Aditya Mohan**, Tome Eftimov, and M. Lindauer. "Instance Selection for Dynamic Algorithm Configuration with Reinforcement Learning: Improving Generalization". In: **GECCO** (2024).
- 7 Alexander Tornede, Difan Deng, Theresa Eimer, Joseph Giovanelli, **Aditya Mohan**, Tim Ruhkopf, Sarah Segel, Daphne Theodorakopoulos, Tanja Tornede, Henning Wachsmuth, and Marius Lindauer. "AutoML in the Age of Large Language Models: Current Challenges, Future Opportunities and Risks". In: **TMLR** (2024).
- 8 **Aditya Mohan***, Carolin Benjamins*, Konrad Wienecke, Alexander Dockhorn, and Marius Lindauer. "AutoRL Hyperparameter Landscapes". In: **Proceedings of the Second International Conference on Automated Machine Learning (AutoML Conf)**. 2023.
- 9 Carolin Benjamins*, Theresa Eimer*, Frederik Schubert, **Aditya Mohan**, Sebastian Döhler, Andre Biedenkapp, Bodo Rosenhahn, Frank Hutter, and Marius Lindauer. "Contextualize Me - The Case for Context in Reinforcement Learning". In: **TMLR** (2023).
- 10 Mohammed Loni*, **Aditya Mohan***, Mehdi Asadi, and Marius Lindauer. "Learning Activation Functions for Sparse Neural Networks". In: **Proceedings of the Second International Conference on Automated Machine Learning (AutoML Conf)**. 2023.
- 11 Tim Ruhkopf, **Aditya Mohan**, Difan Deng, Alexander Tornede, Frank Hutter, and Marius Lindauer. "MASIF: Meta-learned Algorithm Selection using Implicit Fidelity Information". In: **TMLR** (2023).

Preprints & Workshop Papers

- 1 **Aditya Mohan**, Tongle Shen, Fan Feng, Siddhant Agarwal, Caleb Chuck, Benjamin Eysenbach, and Amy Zhang. "Is Manipulation Hard for Successor Measures?" In: **Manuscript, 2026** (Oct. 2026). [URL: https://amsks.github.io/density-fb/](https://amsks.github.io/density-fb/).
- 2 **Aditya Mohan**, Lorenzo Stella, Vihang Patil, Yonatan Naamad, Oleksandr Shchur, and Huibin Shen. "Offline Learning of Flexible Autoscaling Policies from Controller Logs". In: **Manuscript, 2026** (Sept. 2026).
- 3 **Aditya Mohan**, Caleb Chuck, Siddhant Agarwal, and Amy Zhang. "Is Manipulation Hard for Successor Measures?" In: **RLBrew, RLC** (July 2026).
- 4 Dennis Jabs*, **Aditya Mohan***, and Marius Lindauer. "Moments Matter: Stabilizing Policy Optimization using Return Distributions". In: **RLDM**. 2025.
- 5 **Aditya Mohan** and Marius Lindauer. "Towards Enhancing Representations in Reinforcement Learning using Relational Structure". In: **EWRL**. 2024.
- 6 **Aditya Mohan**, Amy Zhang, and Marius Lindauer. "A Patterns Framework for Incorporating Structure in Deep Reinforcement Learning". In: **EWRL**. 2023.

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Aditya Mohan, Tim Ruhkopf, and Marius Lindauer. "Towards Meta-learned Algorithm Selection using Implicit Fidelity Information". In: **ICML 2022 Workshop Adaptive Experimental Design and Active Learning in the Real World (ReALML)**. 2022.

Honors & Awards

- Dec 2024 **Featured Publication in the Binare Magazine for excellent AI research**, *Hannover, Germany*
<https://www.l3s.de/magazine/>
- Oct 2024 **Best Publication Award – L3S**, For our JAIR paper "Structure in Deep Reinforcement Learning: A Survey and Open Problems" *Hannover, Germany*

Funding, Fellowships & Scholarships

- 2025 **DAAD IFI - Internationale Forschungsaufenthalte für Informatikerinnen & Informatiker**, €10,000
- 2020 **EIT Digital Merit Scholarships**, €4,500
- 2017 **DAAD – Praktikenplätze für Ausländische Studierende der Natur- und Ingenieurwissenschaften sowie der Land- und Fortwirtschaft (IAESTE)**, €3,200

Organising

- AI-Connect: Research Meets Industry** *Aachen, Germany*
 ORGANISER Nov 2026
 A mini-workshop on applied problems relevant for researchers
- AutoML School 2024** *Hannover, Germany*
 ORGANISER Sept 2024
- Keynote by Chelsea Finn: Meta-Learning for Education** *Baltimore, MA, USA*
 MODERATOR Jul 2022
- DAC4AutoML Competition at AutoML Conference 2022** *Baltimore, MA, USA*
 ORGANISER Jul 2022

Public Outreach

- Dec 2024 **CAIRNE Rising Researchers Network**, Organizing collaboration between Industry and Academia
- Oct 2024 **Winner – Haus der Wissenschaft Science Slam**, Popular Science Communication Format
- Sep 2024 **Meet the Scientist**, Interacting with School Students about AI
- Sep 2024 **AutoML Conf Non-Traditional Content**, Musical Parody "On the Dangers of Grid Search"
- July 2024 **AI Grid Science Slam**, Popular Science Communication Format, Second Place in Audience Voting
- Nov 2023 **Nacht der Wissenschaft**, University Science Night: RL for all ages

Committees

- Since 2024 **Member**, Appointment Committee for the W3 Databases Professorship at the Faculty of Electrical Engineering and Computer Science *Leibniz Universität Hannover*

Reviewing

JAIR (2024), NeurIPS (2023, 2026), ICML (2022), AutoML Conf (2024, 2023, 2022), EWRL (2024, 2023), ICLR (2024, 2023, 2022), ICLR Tiny Papers (2024, 2023), RLC (2026), TMLR.

Teaching

Oct 2022 - **Reinforcement Learning (3x)**, Graduate lecture: Creation and grading of exercises & final project.
Aug 2025 Teaching concepts for virtual, hybrid, and in-person versions of the course. *Teaching evaluation:* (1.8)
Oct 2024 - **Project: Reinforcement Learning and Robotics**, Graduate level project course: Course
Feb 2025 development & Co-Lecturer
Apr 2022 - **Reinforcement Learning Seminar**, Graduate lecture: Creation and grading of exercises & final
Jul 2022 project. Teaching concepts for virtual, hybrid, and in-person versions of the course

Research Supervision and Mentorship

May 2026 - **Johny Tarbouch (Master Thesis)**, Stitching and Language-Conditioned Compositional
Nov 2026 Generalization
Apr 2025 - **Johns Falkenberg (Bachelor Thesis)**, Dynamics Generalization in Offline Goal-Conditioned RL
Oct 2025
Oct 2024 - **Jan Malte Töpperwein (ML Project, Master Thesis)**, Hyperparameter Landscapes of
Oct 2025 Self-supervised Reinforcement Learning – **PHD Student at LUH Hannover**
Since Apr **Tim Grunwald (ML Project, M.Sc Thesis)**, Prior-fitted Reinforcement Learning for Algorithm
2024 Selection
Feb 2024 - **Dennis Jabs (M.Sc Thesis)**, Improving Policy Optimization Using Return Landscapes – **Accepted at**
Aug 2024 **RLDM 2025**
Feb 2024 - **Dimitrios Timoleon (M.Sc Thesis)**, Enhancing Reinforcement Learning using Transformer-based
Aug 2024 Self-Predictive Representations – **Software Engineer at IAV GmbH**
Since Jun **Wladislaw Petscherski (B.Sc Thesis, ML Project)**, Activation Functions for Transfer-learning in
2023 Reinforcement Learning
Jan 2023 - **Lingxiao Kong (M.Sc Thesis)**, Impact of Hyperparameters on Sim2Real Transfer in Reinforcement
May 2023 Learning – **PHD Student at Fraunhofer FIT**
Oct 2022 - **Konrad Wienecke (M.Sc Thesis)**, Dynamic Hyperparameter Landscapes in Reinforcement
Mar 2023 Learning – **Published at AutoML Conf 2023**